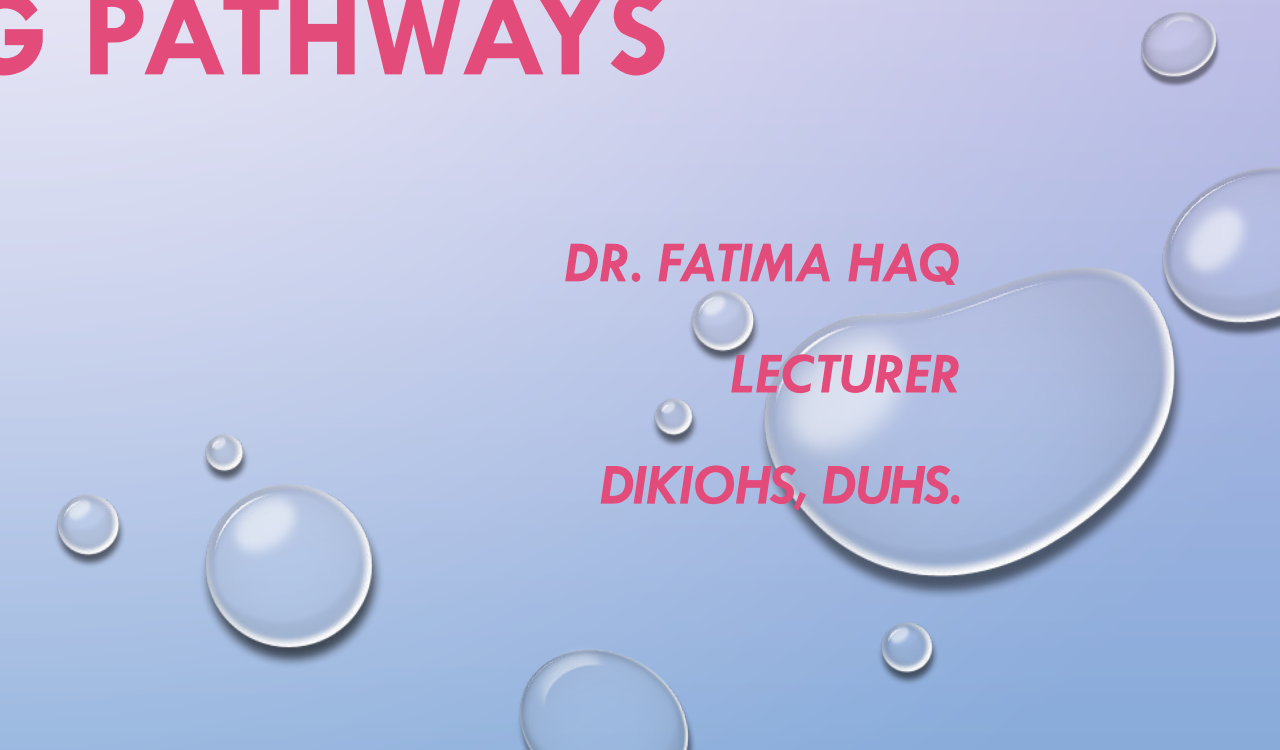




MOTOR CORTEX & ITS DESCENDING PATHWAYS

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LEARNING OBJECTIVES

BY THE END OF THIS LECTURE, STUDENTS SHOULD BE ABLE TO:

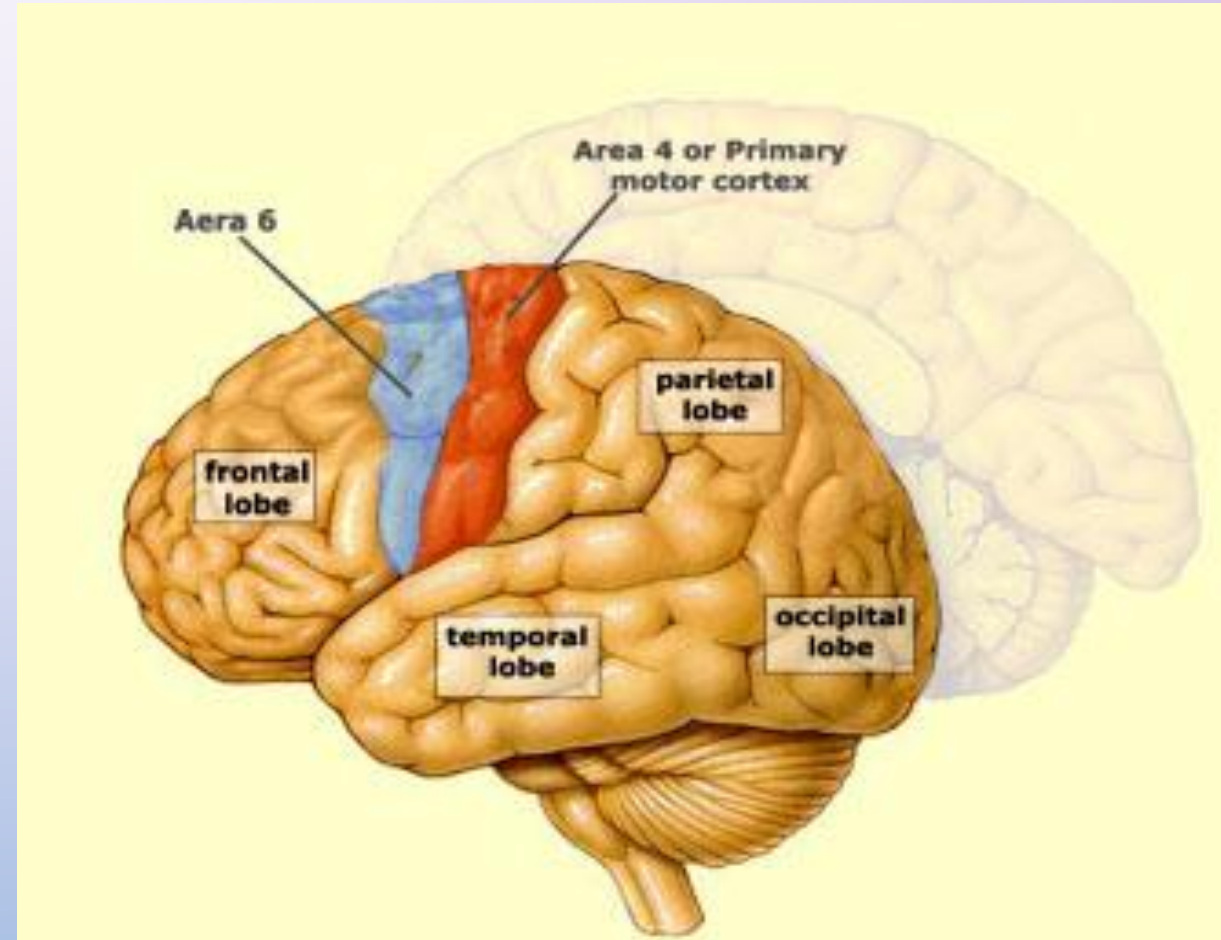
- ✓ DESCRIBE THE ORGANIZATION & FUNCTIONAL AREAS OF THE MOTOR CORTEX INCLUDING PRIMARY MOTOR CORTEX, PREMOTOR CORTEX & SUPPLEMENTARY MOTOR AREA.
- ✓ EXPLAIN THE MAJOR FUNCTIONS OF MOTOR CORTEX IN PLANNING, INITIATION & EXECUTION OF VOLUNTARY MOVEMENTS.
- ✓ EXPLAIN THE CONCEPT OF MOTOR HOMUNCULUS.
- ✓ IDENTIFY THE MAJOR DESCENDING PATHWAYS ARISING FROM CEREBRAL CORTEX I.E. THE CORTICOSPINAL & CORTICOBULBAR PATHWAYS & THEIR FUNCTIONS.
- ✓ IDENTIFY THE MAJOR DESCENDING PATHWAYS OF EXTRAPYRAMIDAL SYSTEM I.E. THE VESTIBULOSPINAL, RETICULOSPINAL, RUBROSPINAL & TECTOSPINAL TRACTS & THEIR FUNCTIONS IN VOLUNTARY MOVEMENT.
- ✓ EXPLAIN THE COMBINED ROLE OF THE MOTOR CORTEX, BASAL GANGLIA & CEREBELLUM IN PLANNING, EXECUTION & CONTROL OF VOLUNTARY MOVEMENTS.

MOTOR CORTEX

- MOTOR CORTEX IS A REGION OF CEREBRAL CORTEX IN THE BRAIN THAT PLANS, CONTROLS & EXECUTES VOLUNTARY MOVEMENTS.

LOCATION

- IT IS LOCATED IN THE FRONTAL LOBE OF THE BRAIN, RIGHT IN FRONT OF A DEEP GROOVE CALLED THE **CENTRAL SULCUS**.
- IT SITS RIGHT AT THE REAR EDGE OF THE FRONTAL LOBE WHERE IT MEETS THE PARIETAL LOBE.



MAIN REGIONS OF MOTOR CORTEX

- THE MOTOR CORTEX SITS ON BOTH SIDES OF THE BRAIN & SPLITS INTO THREE PRIMARY FUNCTIONAL AREAS:

1. PRIMARY MOTOR CORTEX (M1)

- IT IS LOCATED IN THE PRECENTRAL GYRUS OF THE FRONTAL LOBE JUST IN FRONT OF THE CENTRAL SULCUS.
- IT OCCUPIES THE STRIP KNOWN AS THE **PRECENTRAL GYRUS (BRODMANN AREA 4)**.

FUNCTIONS

1. **EXECUTES VOLUNTARY MOVEMENTS:** ACTS AS THE PRIMARY COMMAND CENTER TO TRIGGER & PERFORM SPECIFIC MUSCLES ACTIONS.
2. **CONTROLS FINE MOTOR SKILLS:** GENERATE NEURAL IMPULSES FOR PRECISE, SKILLED MOVEMENTS LIKE USING FINGERS OR FACIAL EXPRESSIONS.
3. **SIGNAL TRANSMISSION:** SENDS MOVEMENT COMMANDS DOWN VIA CORTICOSPINAL & CORTICOBULBAR TRACTS TO ACTIVATE MOTOR NEURONS IN THE SPINAL CORD & BRAIN STEM.

MOTOR HOMUNCULUS

- IT IS A MAP OF HUMAN BODY DRAWN ONTO THE BRAIN'S PRIMARY MOTOR CORTEX.
- PARTS THAT DO FINE, DETAILED MOVEMENTS LIKE HANDS, FINGERS, LIPS & TONGUE ARE HUGE WHILE PARTS THAT DO SIMPLE MOVEMENTS LIKE THE TRUNK & LEGS ARE TINY.
- IT **WORKS CONTRALATERALLY** I.E. THE LEFT SIDE OF THE BRAIN'S MOTOR CORTEX CONTROLS THE RIGHT SIDE OF THE BODY & THE RIGHT SIDE CONTROLS THE LEFT SIDE OF THE BODY.



WHY ARE HANDS & FACE HIGHLY REPRESENTED?

- DIFFERENT BODY REGIONS OCCUPY DIFFERENT SIZE OF CORTICAL AREA.
- THE HANDS, FINGERS, FACE & TONGUE HAVE LARGE CORTICAL REPRESENTATIONS BECAUSE:
 - i. FINE MOTOR CONTROL
 - ii. PRECISE MOVEMENTS
 - iii. COMPLEX PATTERNS OF MOVEMENT
 - iv. INDEPENDENT MOVEMENT OF INDIVIDUAL MUSCLES
- **EXAMPLE:** WRITING REQUIRES MUCH GREATER CORTICAL CONTROL THAN SIMPLY MOVING THE SHOULDER.

2. PREMOTOR CORTEX

- IT IS LOCATED IN THE FRONTAL LOBE OF THE BRAIN JUST ANTERIOR(IN FRONT OF) THE PRIMARY MOTOR CORTEX & POSTERIOR TO THE PREFRONTAL CORTEX.
- IT OCCUPIES THE LATERAL PART OF **BRODMANN AREA 6**.
- IT IS DIVIDED INTO **DORSAL PREMOTOR CORTEX(PMD)** & **VENTRAL PREMOTOR CORTEX(PMV)**.

FUNCTIONS

1. **MOVEMENT PLANNING:** PREPARES & ORGANIZES COMPLEX PATTERNS OF VOLUNTARY MOVEMENT BEFORE ACTUAL EXECUTION.
2. **SENSORY GUIDANCE:** INTEGRATES EXTERNAL SENSORY CUES(LIKE VISUAL TARGETS) TO GUIDE HOW & WHERE TO MOVE.
3. **PROXIMAL CONTROL:** HELPS CONTROL LARGE TRUNK & PROXIMAL MUSCLES USED FOR POSTURE & BODY ORIENTATION.
4. **ACTION UNDERSTANDING:** PARTICIPATES IN MIRROR NEURON SYSTEM TO HELP UNDERSTAND THE ACTIONS PERFORMED BY OTHERS.

3. SUPPLEMENTARY MOTOR AREA (SMA)

- IT IS LOCATED ON THE MEDIAL(INNER) SURFACE OF THE HEMISPHERE NEAR THE TOP OF THE BRAIN.
- IT IS POSITIONED JUST IN FRONT OF THE PRIMARY MOTOR CORTEX & ON TOP OF(SUPERIOR TO) THE PREMOTOR AREA.
- IT IS DIVIDED INTO CAUDAL PART CALLED **SMA PROPER** & ROSTRAL PART CALLED **PRE SMA**.

FUNCTIONS

1. **INTERNAL PLANNING:** CONTROL MOVEMENTS THAT ARE GENERATED FROM MEMORY OR THOUGHT RATHER THAN REACTING TO SUDDEN EXTERNAL SIGHTS OR THOUGHTS.
2. **COORDINATION:** HELPS MANAGE MOVEMENTS THAT USE BOTH SIDES OF THE BODY AT THE SAME TIME SUCH AS PLAYING A PIANO OR TYPING.



3. SEQUENCE CONTROL: ORGANIZES A SMOOTH CHAIN OF MULTIPLE MOVEMENTS IN THE CORRECT ORDER.

4. POSTURAL STABILIZATION: HELPS KEEP THE BODY STABLE & BALANCED DURING ACTION.

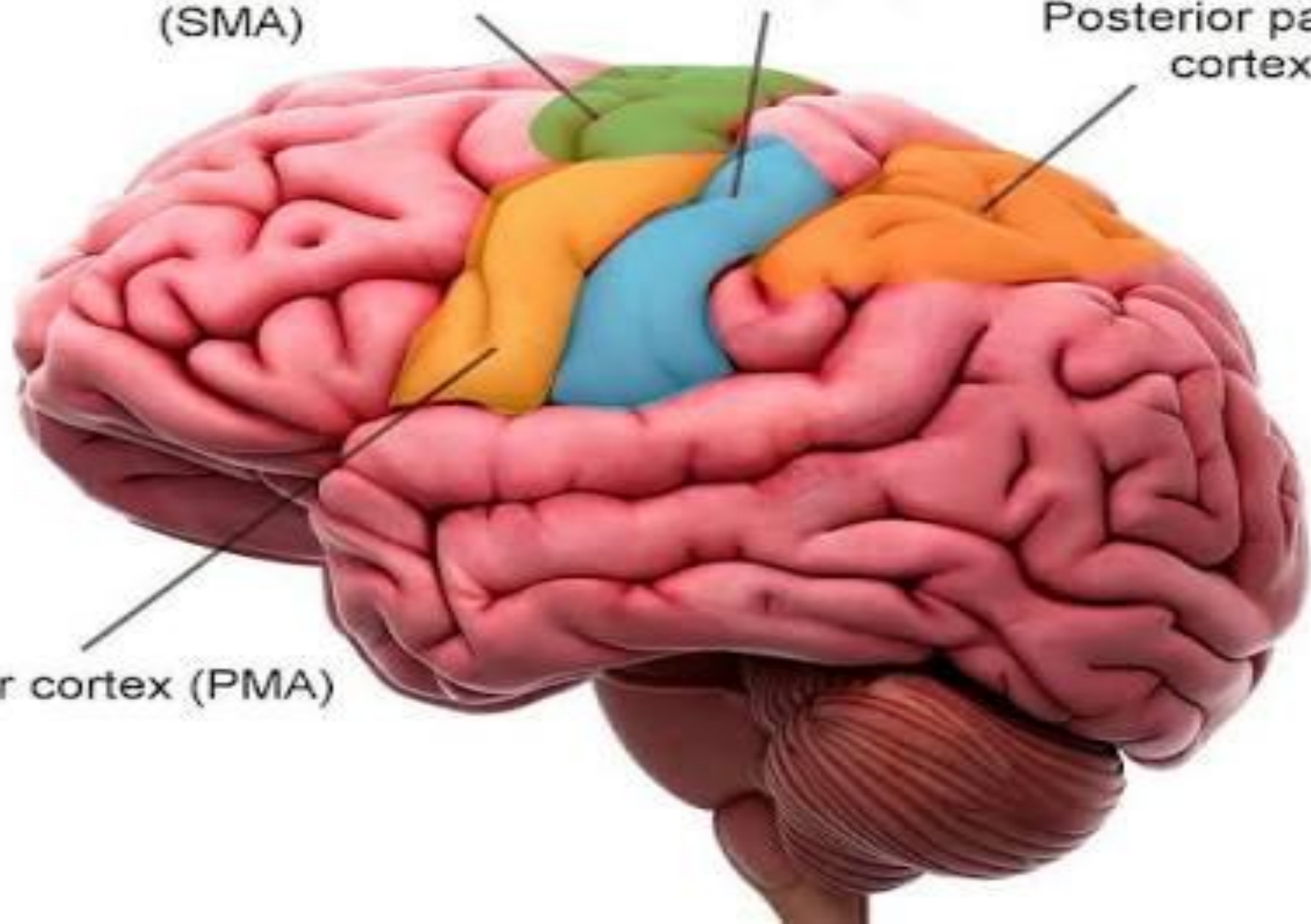


Supplementary
motor cortex
(SMA)

Primary motor
cortex (M1)

Posterior parietal
cortex

Premotor cortex (PMA)



MOTOR CORTEX INPUTS

□ THE MOTOR CORTEX RECEIVES INFORMATION FROM SEVERAL REGIONS, MAJOR INPUTS TO MOTOR CORTEX INCLUDE:

- i. **SOMATOSENSORY CORTEX:** LOCATED RIGHT BEHIND THE MOTOR CORTEX & SENDS DIRECT FEEDBACK ABOUT BODY POSITION & TOUCH.
- ii. **CEREBELLUM:** IT SENDS SIGNALS THROUGH THE VENTRAL LATERAL (VL) NUCLEUS OF THE THALAMUS TO HELP WITH BALANCE, TIMING & SMOOTH MOVEMENT.
- iii. **BASAL GANGLIA:** SENDS SIGNALS THROUGH THE THALAMUS TO HELP SELECT & CONTROL VOLUNTARY MOVEMENTS.
- iv. **THALAMUS(DIRECT SENSORY):** RECEIVES TOUCH & POSITION SIGNALS FROM THE BODY & PASSES THEM STRAIGHT TO THE MOTOR CORTEX.
- v. **OTHER CORTICAL AREAS:** THE PRIMARY MOTOR CORTEX GETS DIRECT SIGNALS FROM THE PREMOTOR CORTEX & THE SUPPLEMENTARY MOTOR AREA WHICH HELP PLAN & COORDINATE MOVEMENTS.

MOTOR CORTEX OUTPUTS

□ THE MOTOR CORTEX SENDS SIGNALS THROUGH MAJOR DESCENDING PATHWAYS TO CONTROL VOLUNTARY MOVEMENTS OF THE BODY, HEAD, FACE & NECK.

1. **PYRAMIDAL SYSTEM OR PATHWAYS:** INCLUDES **CORTICOSPINAL** & **CORTICOBULBAR PATHWAYS**.
 2. **EXTRAPYRAMIDAL SYSTEM OR PATHWAYS:** DESCENDING PATHWAYS FROM THE BRAINSTEM I.E. THE **RUBROSPINAL**, **RETICULOSPINAL**, **VESTIBULOSPINAL** & **TECTOSPINAL** PATHWAYS ARE GROUPED UNDER EXTRAPYRAMIDAL PATHWAYS.
- THESE SYSTEMS WORK TOGETHER TO CONTROL VOLUNTARY MOVEMENT, POSTURE & MUSCLE TONE.

Pyramidal Tract

Originated from Brain cortex
Responsible for voluntary movement.

Has 2 tracts :

Corticospinal Tract:
for skeletal muscle voluntary movement

25 % Ipsilateral , 75%contralateral

Corticobulbar tract :
for face, head, neck voluntary movement.
innervate both sides equally except for facial nerve (7th) supply lower third of contralateral side of face



Extrapyrarnidal Tract

Originated from Brainstem
Responsible for involuntary movement.

Has 4 tracts :

Rubrospinal Tract
for Fine movement control

Reticulospinal Tract
For tone control and automate breathing

Vistebulospinal Tract
For Balance and posture.

Tectospinal Tract
for head- eye- body cordination



CORTICOCORTICAL & SUBCORTICAL OUTPUTS

FEEDBACK LOOPS TO THE CEREBELLUM

- SENDS AXONS COLLATERALS THROUGH PONTINE NUCLEI BACK TO THE CEREBELLUM TO HELP COORDINATE & LEARN MOTOR SKILLS.

PROJECTIONS TO SUBCORTICAL NUCLEI

- SENDS FIBERS TO THE BASAL GANGLIA(SUCH AS PUTAMEN), THE RED NUCLEUS & THE RETICULAR FORMATION TO REFINE & REGULATE MOTOR OUTPUT.

CORTICOSPINAL TRACT

- THE CORTICOSPINAL TRACT IS THE MAIN NERVE PATHWAY THAT CARRIES VOLUNTARY MOVEMENT SIGNALS FROM THE BRAIN DOWN TO THE BODY MUSCLES.
- IT STARTS IN THE BRAIN'S MOTOR CORTEX, TRAVELS THROUGH THE BRAINSTEM & CONNECTS TO THE SPINAL NERVES THAT **CONTROL THE ARMS, LEGS & TRUNK**.
- THE CORTICOSPINAL PATHWAY IS PARTICULARLY IMPORTANT FOR SKILLED, PRECISE VOLUNTARY MOVEMENTS ESPECIALLY OF HANDS & FINGERS.

WHERE THE PATHWAY BEGINS?

MOTOR CORTEX: SIGNALS START IN THE HIGHER CENTERS OF THE BRAIN THAT PLANS & STARTS MOVEMENT.

UPPER MOTOR NEURONS: THESE NERVE CELLS SEND MESSAGES DOWN FROM THE BRAIN.

BRAINSTEM LEVEL: THE NERVE FIBERS PASS TIGHTLY TOGETHER THROUGH DEEP BRAIN AREAS SUCH AS THE INTERNAL CAPSULE & THE BRAINSTEM PYRAMIDS.

HOW THE TRACT DIVIDES?

DECUSSATION(CROSSING OVER)

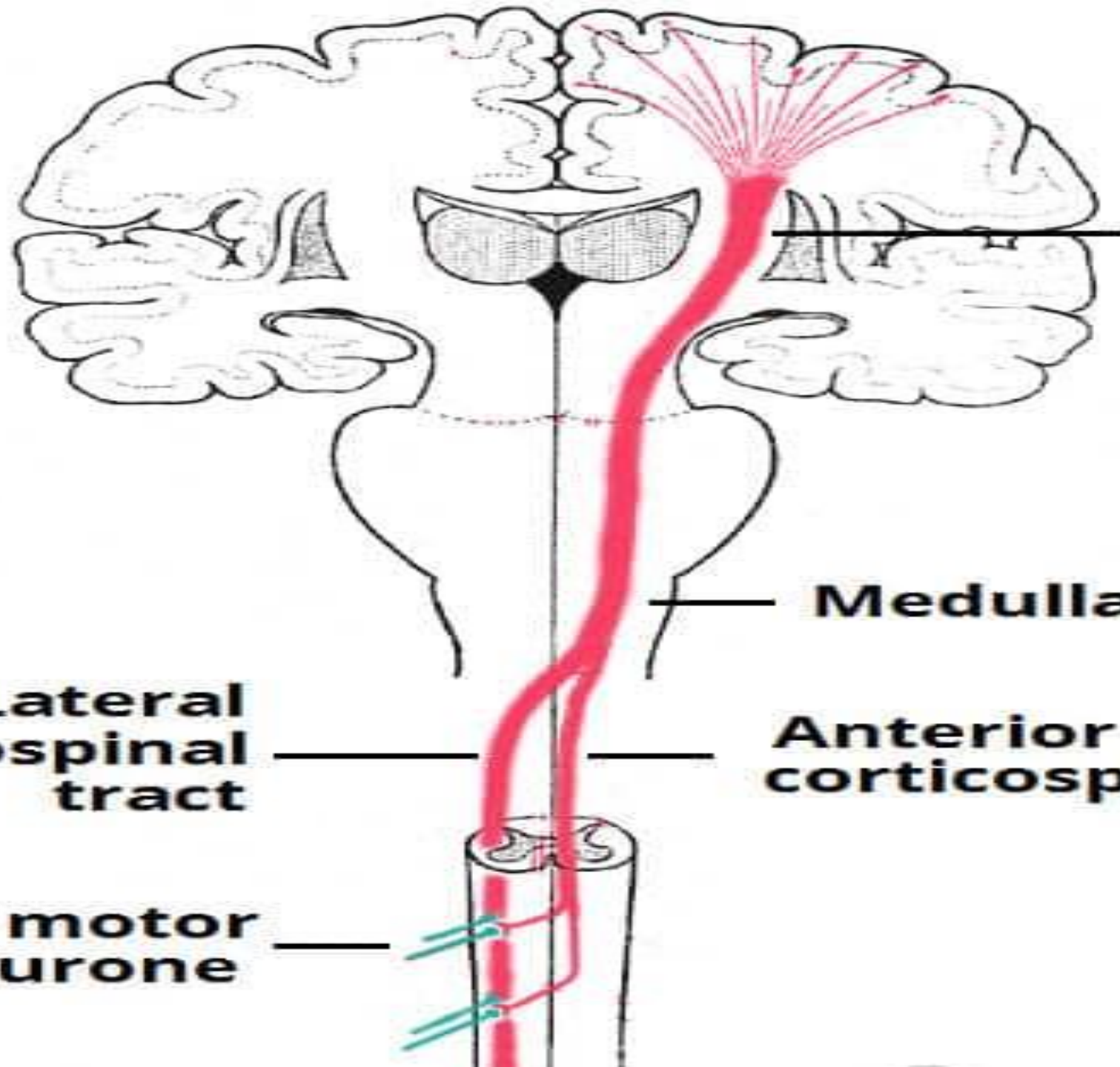
- NEAR THE BOTTOM OF THE BRAINSTEM, ABOUT 90% OF THE NERVE FIBERS CROSS TO THE OPPOSITE.

LATERAL CORTICOSPINAL TRACT

- DESCENDS IN THE LATERAL WHITE MATTER OF SPINAL CORD.
- THE FIBERS THAT CROSS CONTROL PRECISE SKILLED MOVEMENTS IN THE DISTANT MUSCLES OF THE BODY(LIKE FINGERS & TOES).

ANTERIOR CORTICOSPINAL TRACT

- DESCENDS MAINLY IN THE ANTERIOR SPINAL CORD.
- THE SMALL GROUP OF FIBERS(ABOUT 10%) THAT DO NOT CROSS & STAY ON THE SAME SIDE TO HELP CONTROL TRUNK & POSTURE MUSCLES.



Internal capsule

Medulla

Lateral corticospinal tract

Anterior corticospinal tract

Lower motor neurone

FINAL STEPS TO MOVE MUSCLES?

LOWER MOTOR NEURONS

- UPPER MOTOR NEURONS HOOK UP WITH LOWER MOTOR NEURONS INSIDE THE SPINAL CORD.

MUSCLE ACTION

- LOWER MOTOR NEURONS CARRY THE FINAL SIGNAL DIRECTLY TO THE SKELETAL MUSCLES TO MAKE THEM SQUEEZE OR RELAX.

CORTICOBULBAR TRACT

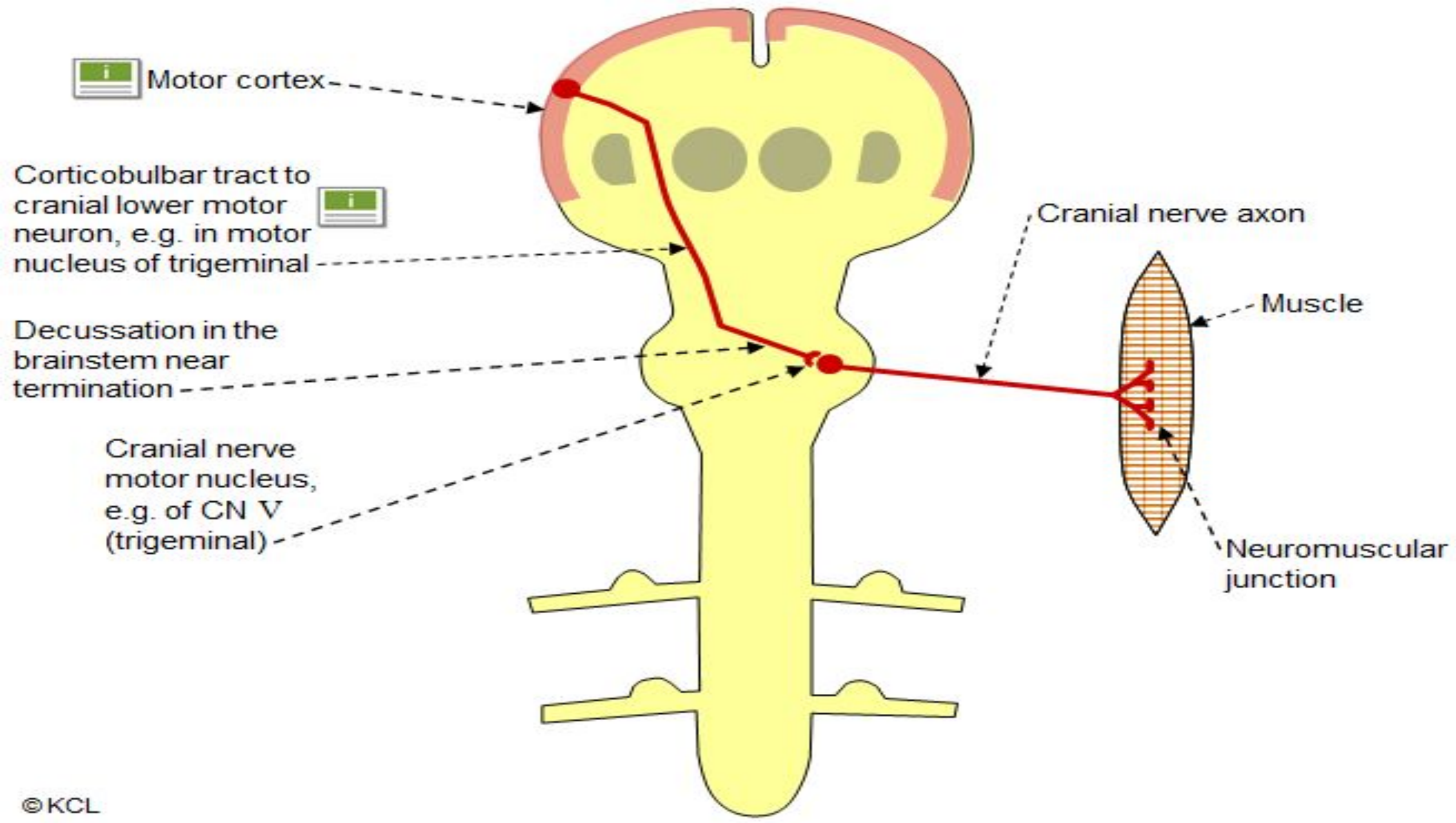
- IT IS AN **UPPER MOTOR NEURON PATHWAY** THAT SEND MOTOR COMMANDS FROM THE CEREBRAL CORTEX TO MOTOR NUCLEI OF CRANIAL NERVES.
- IT CONTROL **VOLUNTARY MOVEMENTS OF THE FACE, HEAD & NECK**.
- CORTICOBULBAR TRACT IS PARTICULARLY IMPORTANT FOR FACIAL EXPRESSIONS, CHEWING, SWALLOWING, SPEECH & TONGUE MOVEMENTS AS IT CONTROLS MOVEMENTS INVOLVING FACIAL MUSCLES, MUSCLES OF MASTICATION, TONGUE, PHARYNX & LARYNX.

BILATERAL INNERVATION

- MOST CRANIAL NERVE NUCLEI SUCH AS **V, VII(UPPER PORTION), IX, X, XI** RECEIVE SIGNALS FROM BOTH THE RIGHT & LEFT MOTOR CORTEX.

CONTRALATERAL INNERVATION

- SOME OF THE LOWER MOTOR NUCLEI SUCH AS **LOWER PORTION OF FACIAL NUCLEUS (CONTROLLING LOWER FACIAL EXPRESSIONS) & PART OF HYPOGLOSSAL NUCLEUS (CONTROLLING TONGUE DEVIATION)** RECEIVE PRIMARILY CONTRALATERAL (OPPOSITE SIDE) INPUT.



BRAINSTEM DESCENDING PATHWAYS

- BRAINSTEM DESCENDING PATHWAYS ARE THE NEURAL TRACTS THAT ORIGINATE IN THE BRAINSTEM & CARRY MOTOR COMMANDS DOWN TO THE SPINAL CORD TO **CONTROL BALANCE, POSTURE & AUTOMATIC MOVEMENTS**.

MAJOR FUNCTIONS

- INVOLUNTARY CONTROL:** CONTROLS UNCONSCIOUS BODY ADJUSTMENTS LIKE MUSCLE TONE & EQUILIBRIUM.
- REFLEX COORDINATION:** WORKS WITH SENSORY INPUTS TO REACT QUICKLY TO ENVIRONMENTAL CHANGES.
- VOLUNTARY SUPPORT:** PREPARES BODY'S POSTURE BEFORE INTENTIONAL MOVEMENTS OCCUR.

MAJOR TRACTS

1. **VESTIBULOSPINAL TRACT:** CONTROLS BALANCE & POSTURE .
2. **RETICULOSPINAL TRACT:** REGULATES MUSCLE TONE, REFLEX ACTIVITY & AUTOMATIC BREATHING.
3. **RUBROSPINAL TRACT:** FACILITATES ACTIVITY OF FLEXOR MUSCLES & ASSISTS WITH FINE MOTOR CONTROL.
4. **TECTOSPINAL TRACT:** COORDINATES HEAD & EYE MOVEMENTS IN RESPONSE TO VISUAL CUES.

VESTIBULOSPINAL TRACT

- IT IS A GROUP OF NERVE FIBERS IN THE SPINAL CORD THAT HELPS TO **KEEP BALANCE & STAND UPRIGHT**.
- IT SENDS SIGNALS FROM THE INNER EAR & BRAINSTEM TO THE MUSCLES TO ADJUST POSTURE WHILE MOVING.

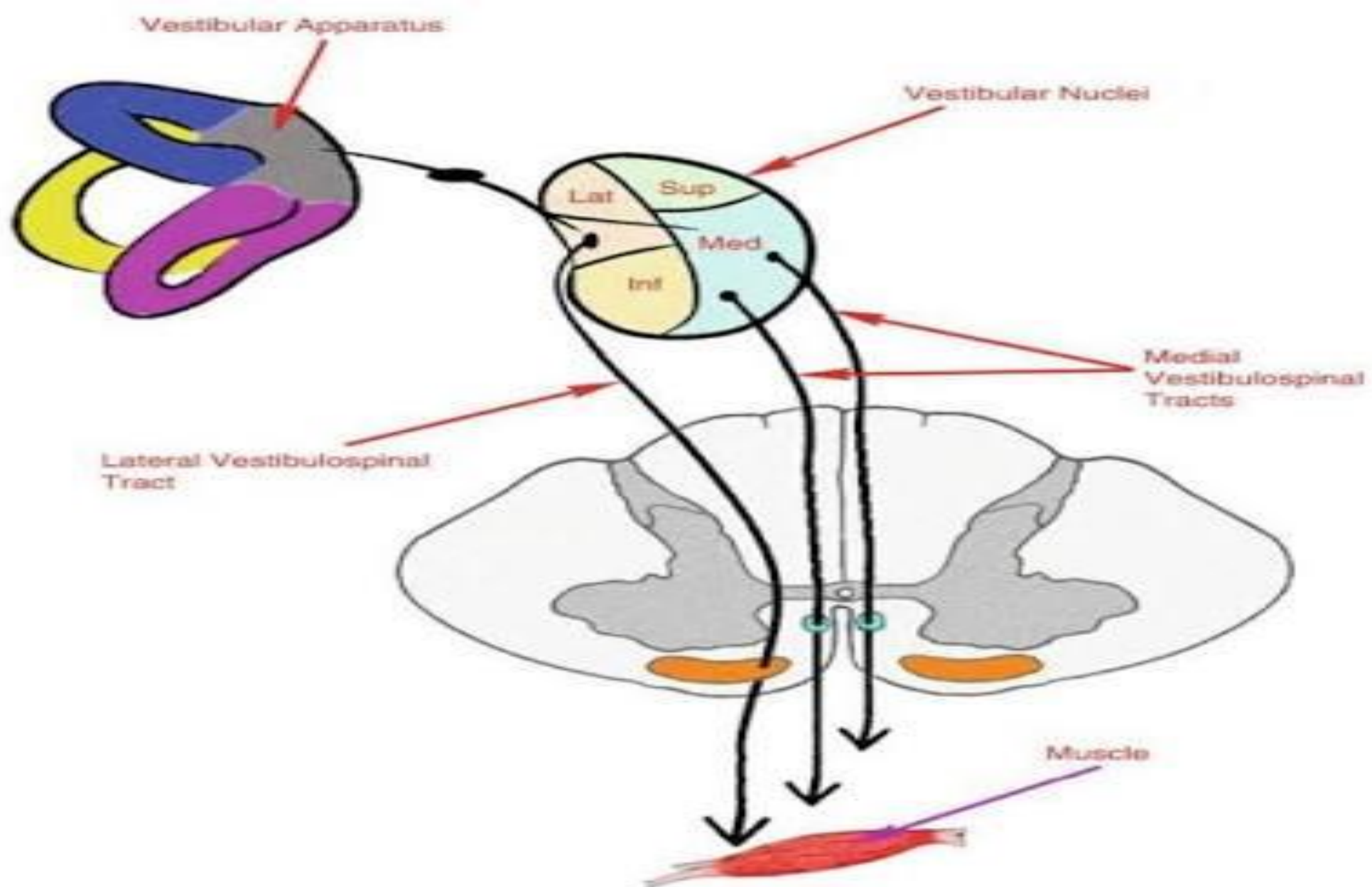
MAIN PARTS OF VESTIBULOSPINAL TRACT

LATERAL TRACT

- IT STARTS IN THE LATERAL BRAINSTEM & CONTROLS MUSCLES IN THE LEGS & TRUNK TO PROTECT FROM FALLING.

MEDIAL TRACT

- IT STARTS NEAR THE MIDDLE OF THE BRAINSTEM & HELPS CONTROL MUSCLES IN THE NECK & UPPER BACK TO KEEP THE HEAD STEADY.



MAJOR FUNCTIONS OF VESTIBULOSPINAL TRACT

1. *BALANCE*

- THE VESTIBULOSPINAL TRACT RESPONDS FAST WHEN THE HEAD TILTS OR MOVES.

2. *MUSCLE TONE*

- THE VESTIBULOSPINAL TRACT ACTIVATES ANTI-GRAVITY(EXTENSORS) MUSCLES TO STAY UPRIGHT(PROTECT FROM FALLING).

3. *COORDINATION*

- VESTIBULOSPINAL TRACT WORKS WITH THE EYES & INNER EAR TO KEEP VISION CLEAR WHILE HEAD MOVES.

RETICULOSPINAL TRACT

- ❑ RETICULOSPINAL TRACT IS A MAJOR DESCENDING PATHWAY THAT STARTS IN THE BRAINSTEM.
- ❑ IT HELPS CONTROL POSTURE, BALANCE & WALKING MOVEMENTS BY MANAGING MUSCLE TONE & BODY POSITION.

MAIN PARTS OF RETICULOSPINAL TRACT

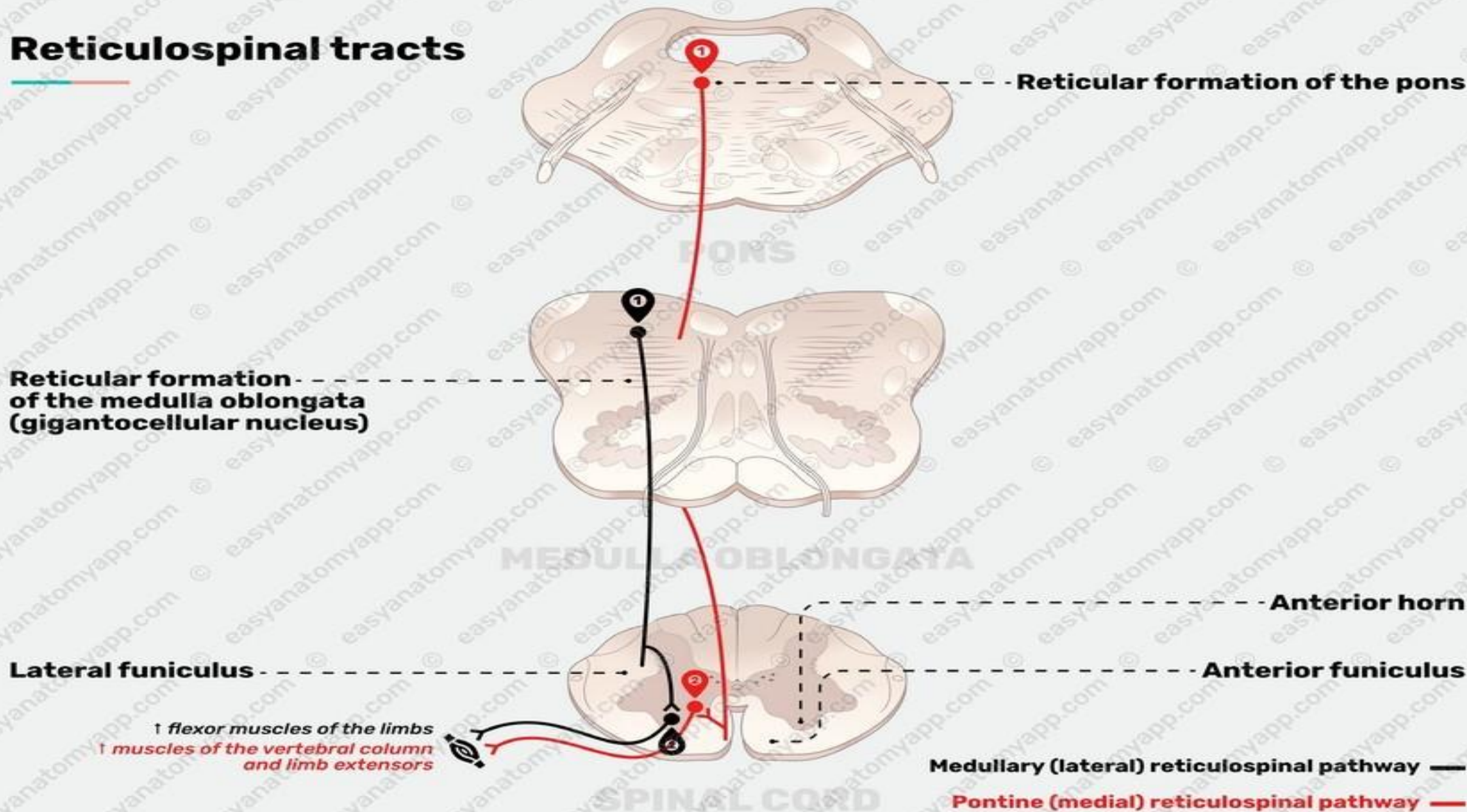
MEDIAL(PONTINE) TRACT

- ❑ IT STARTS IN THE PONS & EXCITES EXTENSOR MUSCLES TO HELP STAND AGAINST THE GRAVITY.

LATERAL(MEDULLARY) TRACT

- ❑ STARTS IN THE MEDULLA & CAUSE INHIBITION OF THE EXTENSOR MUSCLES.
- ❑ THIS TRACT HELPS CONTROL BASIC BODY MOVEMENTS.

Reticulospinal tracts



MAJOR FUNCTIONS OF RETICULOSPINAL TRACT

1. *POSTURE CONTROL*

□ IT HELPS KEEP THE BODY STABLE WHILE MOVING.

2. *MUSCLE TONE*

□ THE RETICULOSPINAL TRACT ADJUST THE TIGHTNESS(TONE) OF MUSCLES AT REST POSITION.

3. *MOVEMENT SUPPORT*

□ THIS TRACT ASSISTS BASIC MOTION LIKE WALKING.

RUBROSPINAL TRACT

- THE RUBROSPINAL TRACT STARTS IN THE RED NUCLEUS SITUATED IN THE MIDBRAIN.
- JUST AFTER LEAVING THE RED NUCLEUS, THE NERVE FIBERS CROSS TO THE OPPOSITE SIDE OF THE BRAIN(DECUSSATION).
- THE FIBERS THEN TRAVEL DOWN THE SIDE OF THE BRAINSTEM TO ENTER THYE LATERAL PART OF THE SPINAL CORD.
- FINALLY, THE NERVE FIBERS CONNECT WITH MOTOR NEURONS IN THE SPINAL CORD THAT CONTROL MUSCLES.

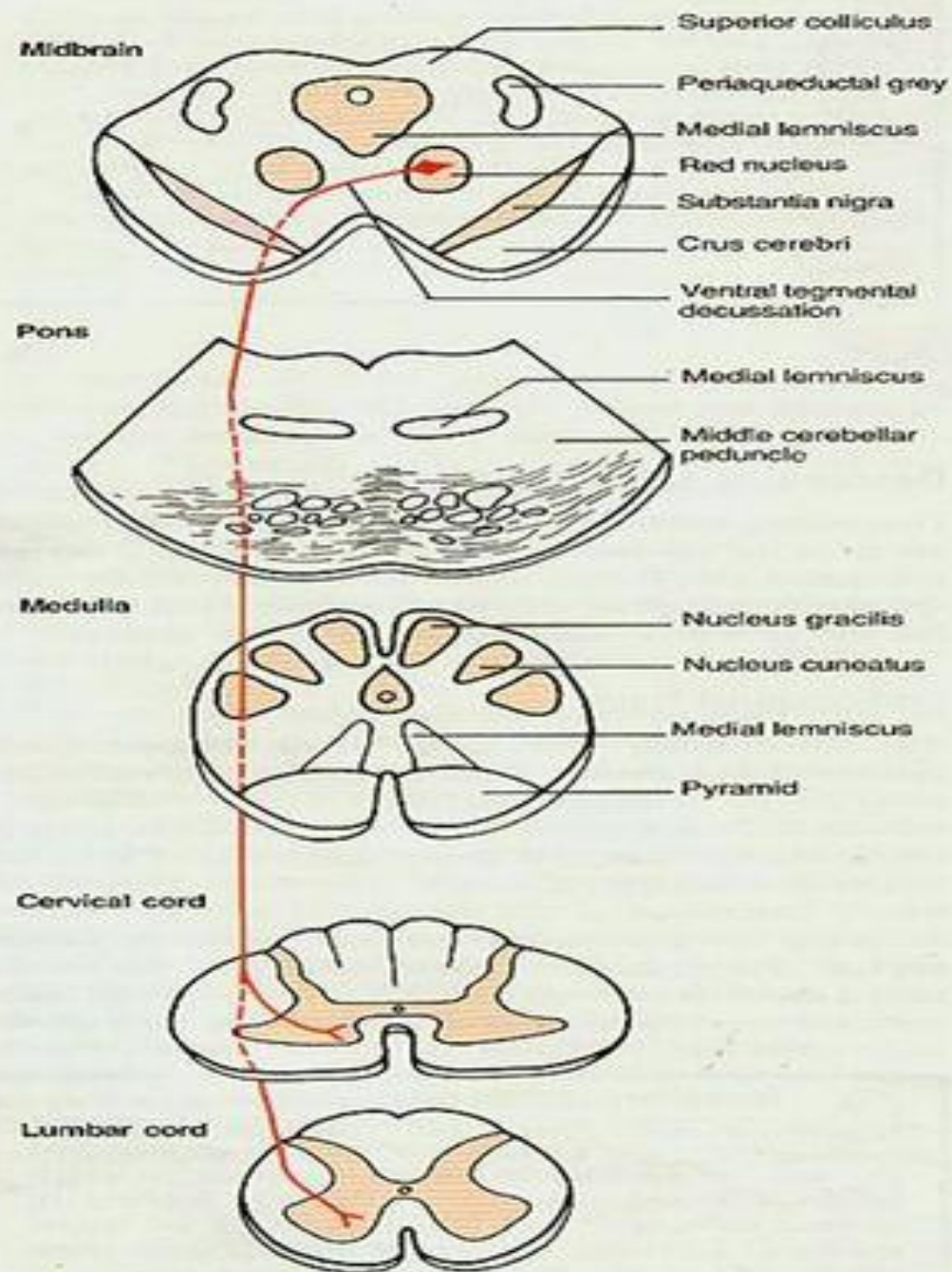
Rubrospinal Tract

- Originate from nerve cells in red nucleus

nerve fibres / axons

- cross the mid line
- descend as rubrospinal tract
 - through pons and medulla oblongata
- Terminate in the anterior grey column of spinal cord

(facilitate the activity of flexor muscles)



MAJOR FUNCTIONS OF RUBROSPINAL TRACT

1. FLEXOR STIMULATION

- THE RUBROSPINAL TRACT EXCITES THE FLEXOR MUSCLES THAT BEND THE LIMBS ESPECIALLY IN THE UPPER ARMS & HANDS.

2. EXTENSOR INHIBITION

- THIS TRACT INHIBITS THE EXTENSOR MUSCLES THAT STRAIGHTEN THE LIMBS.

3. MUSCLE TONE

- THIS TRACT ALSO MAINTAIN STEADY MUSCLE TENSION FOR BASIC MOVEMENTS.

TECTOSPINAL TRACT

- THE TECTOSPINAL TRACT IS A MAJOR DESCENDING PATHWAY THAT STARTS IN THE SUPERIOR COLLICULUS OF THE MIDBRAIN(TECTUM).
- FIBERS THEN CROSS TO THE OPPOSITE SIDE OF THE BRAIN(DECUSSATION).
- THE NERVE FIBERS THEN TRAVEL DOWN TO ENTER THE CERVICAL PARTS OF SPINAL CORD.

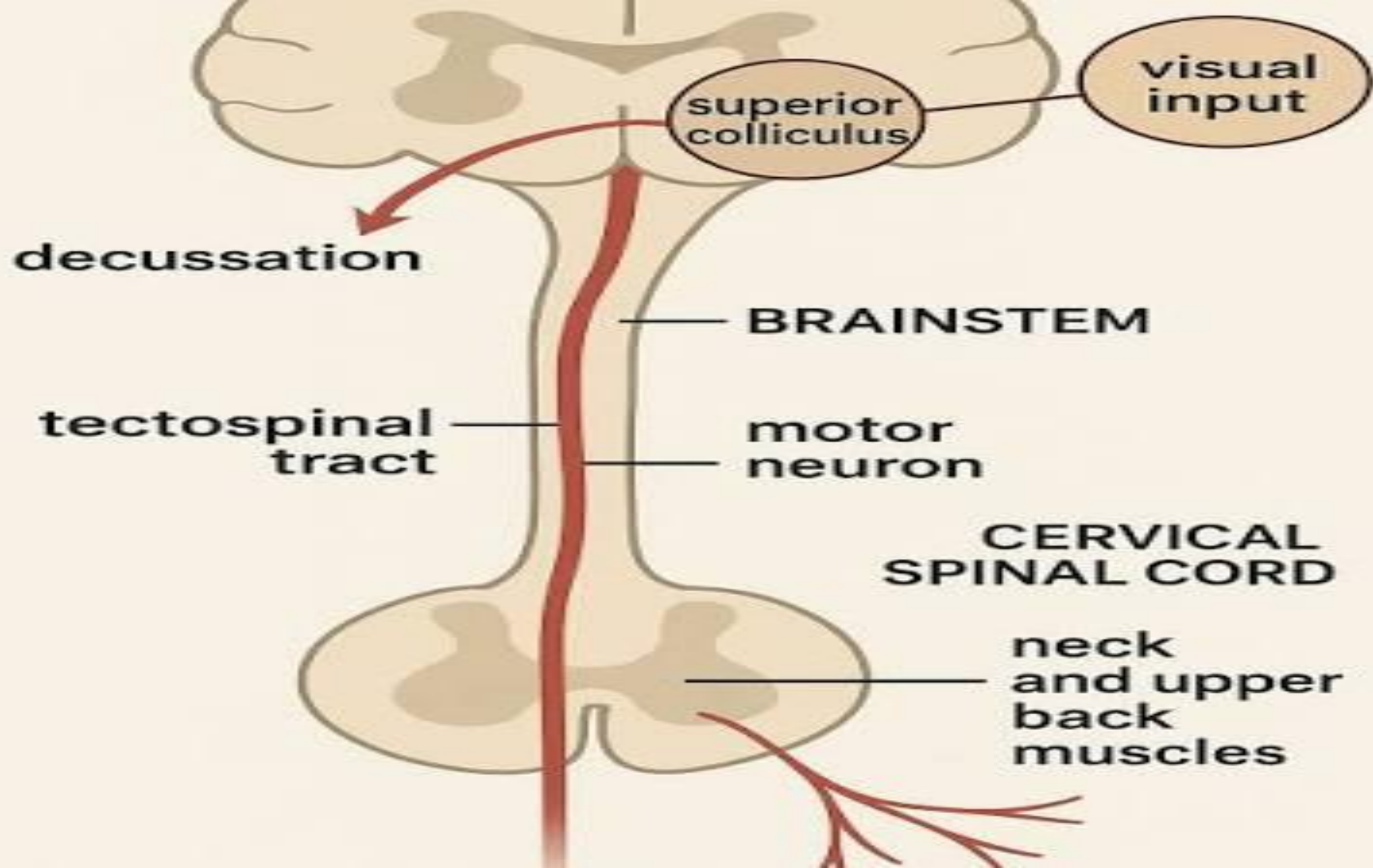
MAJOR FUNCTIONS OF TECTOSPINAL TRACT

1. REFLEX CONTROL

- THE TECTOSPINAL TRACT DIRECTS HEAD & EYES TOWARDS SIGHTS & SOUNDS.

2. POSTURE CONTROL

- THIS TRACT ALSO MANAGES NECK MUSCLES TO KEEP BALANCE & TO FACE A NEW STIMULUS.



MOTOR CORTEX & VOLUNTARY MOVEMENTS

- THE MOTOR CORTEX PLANS, CONTROLS & EXECUTES ALL VOLUNTARY MUSCLE MOVEMENTS.
- THE LEFT MOTOR CORTEX CONTROLS THE RIGHT SIDE OF THE BODY WHEREAS THE RIGHT MOTOR CORTEX CONTROLS THE LEFT SIDE OF THE BODY (**CONTRALATERAL CONTROL**).
- THE SIGNALS FROM THE MOTOR CORTEX THEN TRAVELS DOWN VIA CORTICOSPINAL TRACT FOR THE BODY MOVEMENTS & VIA CORTICOBULBAR TRACT FOR THE MOVEMENTS OF FACE & HEAD.
- BODY PARTS ARE MAPPED ACROSS THE CORTEX PROPORTIONALLY TO THE PRECISION OF MOVEMENTS (**MOTOR HOMUNCULUS**) I.E. LARGER AREAS ARE GIVEN TO HANDS & LIPS THAN TRUNK & LEGS.

REGIONS OF MOTOR CORTEX & THEIR RELATIVE FUNCTIONS

PRIMARY MOTOR CORTEX(M1)

- PRIMARY MOTOR CORTEX EXECUTES ACTUAL MOVEMENTS BY SENDING DIRECT ELECTRICAL SIGNALS DOWN TO THE SPINAL CORD & MUSCLES.

PREMOTOR CORTEX

- PREMOTOR CORTEX HELPS TO ASSEMBLE MOTOR PROGRAM I.E. PREPARES & PLANS MOVEMENT BEFORE THE EXECUTION.

SUPPLEMENTARY MOTOR AREA(SMA)

- IT MANAGES & CONTROLS COMPLEX SEQUENCE OF MOVEMENTS LIKE TYPING, PLAYING A PIANO.

COMBINED ROLE OF MOTOR CORTEX, BASAL GANGLIA & CEREBELLUM IN VOLUNTARY MOVEMENTS

- THE MOTOR CORTEX, BASAL GANGLIA & CEREBELLUM WORK TOGETHER IN A FEEDBACK LOOP TO PLAN, INITIATE & REFINE VOLUNTARY MOVEMENTS.
- THE MOTOR CORTEX GENERATES & SENDS THE MAIN MOTOR SIGNALS DOWN THE SPINAL CORD TO ACTIVATE SPECIFIC MUSCLES & EXECUTE THE PHYSICAL MOVEMENT.
- THE BASAL GANGLIA DECIDES WHEN TO START A MOVEMENT & ACTIVELY SUPPRESSES THE COMPETING OR UNWANTED MOVEMENTS VIA DIRECT & INDIRECT PATHWAYS.
- HENCE, THE BASAL GANGLIA ACTS AS AN EDITOR TO ENSURE ONLY THE DESIRED MOTOR PLAN GETS EXECUTED.
- THE CEREBELLUM COMPARES THE INTENDED MOVEMENT FROM THE MOTOR CORTEX WITH ACTUAL SENSORY FEEDBACK FROM MUSCLES & MAKES INSTANT ADJUSTMENTS TO CORRECT ANY ERROR.
- THEREFORE, THE CEREBELLUM ENSURES THE SMOOTHNESS, TIMING & EXECUTION IN CORRECT SEQUENCE.

POST ASSESSMENT

Q1. THE PRIMARY MOTOR CORTEX CORRESPONDS APPROXIMATELY TO WHICH BRODMANN AREA?

- a) AREA 3
- b) AREA 4
- c) AREA 6
- d) AREA 17

Q2. WHICH CORTICAL REGION IS MOST INVOLVED IN PLANNING & PROGRAMMING COMPLEX VOLUNTARY MOVEMENTS?

- a) PRIMARY MOTOR CORTEX
- b) PRIMARY VISUAL CORTEX
- c) PRIMARY AUDITORY CORTEX
- d) PREMOTOR & SUPPLEMENTARY MOTOR AREA

CONT..

Q3. WHICH BODY REGION GENERALLY HAS A DISPROPORTIONATELY LARGE REPRESENTATION IN THE PRIMARY MOTOR CORTEX?

- a) TRUNK
- b) FOOT
- c) HANDS & FINGERS
- d) BACK

Q4. THE CORTICOBULBAR TRACT PRIMARILY CONTROLS:

- a) MUSCLES OF FACE, JAW, TONGUE & PHARYNX
- b) MUSCLES OF LIMBS
- c) MUSCLES INVOLVED IN EYE ACCOMMODATION
- d) MUSCLES OF INTESTINE